

MATTHEW SZARMACH

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OBJECTIVE

Computer Science graduate with a strong foundation in programming and problem-solving, eager to apply my skills in software development. I am passionate about building efficient, scalable solutions and excited to contribute to real-world projects that drive organizational success.

EDUCATION

Bachelor of Computer Science , Pennsylvania State University	2025
Associate of Biomedical Engineering Technology , Pennsylvania State University	2021

PROJECTS

Skeletal Age Estimation (<https://github.com/Skeletal-Bone-Estimation/Skeletal-Age-Estimation>)

- Developed a cross-Platform desktop application leveraging the Electron framework for Mercyhurst's Applied Forensic Science Department.
- Application receives bone marker scoring of skeletal structures and calculates the age of the skeleton case.
- Designed the user interface utilizing the Bootstrap framework and custom CSS.
- Created testing suites leveraging the testing framework Jest.
- Implemented the software design pattern Model-View-Controller (MVC) to manage the application's data flow.
- Incorporated machine learning by experimenting with models, training them on skeletal images to determine the age range of the current case.

LevelUp

- Developed a website that utilizes Steam's Web APIs to allow users to track games they wish to achievement hunt.
- Leveraged Postman to ensure website's API functionality was operating correctly.
- Incorporated MongoDB to manage user information such as LevelUp account, steamID, games tracked, and gaming platform.
- Applied the Bootstrap framework to design the user interface of the website.

Cross Species Emotion Detection

- Developed a Python application that uses a machine learning model to analyze emotion based off an image of a human or dog's face.
- Implemented the convolutional neural network architecture VGG-16 as the model and acquired over 10,000 training images from the website Kaggle.
- Utilized Python libraries TensorFlow, Keras, Matplotlib, NumPy, Seaborn, and Scikit-learn for training and analysis, and PyQt6 for creating the user interface of the app.

COURSEWORK AND SKILLS

Coursework: Artificial Intelligence, Engineering Design, Digital Electronics, Databases, Machine Learning, Web-services, Data Structures and Algorithms

Programming Languages and Data Types: C, C++, C Sharp, Java, Assembly, Python, HTML, CSS, JavaScript, Typescript, R, SAS, Rust, JSON, XML

Frameworks: Electron, React Native, Bootstrap, Avalonia, Ratatui, .NET, Jest

Applications and Tools: Node.js, Docker, Postman, MongoDB, Microsoft SQL Server, Git, GitHub.

WORK EXPERIENCE

Help Desk Technician

Plum Borough School District

Jun 2025 - Present

Plum, PA

- Provided technical support and device repairs for over 1,000 students, faculty, and staff across two schools.
- Communicated effectively through a ticketing system and in-person interactions to ensure timely issue resolution.
- Collaborated with team members to configure and deploy device scripts and profiles for optimal functionality.

Student Research Assistant

Virtual/Augmented Reality (VAR) Lab

Oct 2023 - May 2025

Erie, PA

- Developed mobile applications using the React Native framework.
- Managed the entire software development lifecycle, including building, testing, and debugging augmented reality applications.
- Worked independently and collaboratively in team environments to achieve project goals.
- Integrated microcomputers and microcontrollers (Raspberry Pi and Arduino) into current and past projects.
- High resolution 3D models captured using Atrec 3D technology.
- Engaged with Virtual Reality environments for research and development purposes.

ACTIVITIES

- Robotics Team; Green Roof Project Team Leader
- **Advent of Code 2024:** yearly advent calendar of coding challenges that can be solved in any programming language for a variety of skill levels.

PUBLICATIONS AND PRESENTATIONS

- Pendergass, W., Compomizzi, J. Scibelli, D. and Szarmach, M. (2022). Digital Mandalas: Communication and Authentic Human Interaction in Reddit's r/place Platform. Issues in Information Systems, 2022, Vol.23, Issue 3, pp. 9-24
- Pendergass, W., Compomizzi, J. Scibelli, D. and Szarmach, M. (2022).. International Association for Computer Information Systems Conference. Digital Mandalas: Communication and Authentic Human Interaction in Reddit's r/place Platform Presentation